

# Shift Scheduler – Complete Setup Guide

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## What Was Built

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A **Flask web application** that automatically generates monthly shift schedules from your existing employee roster, fully accessible from any Android browser or PC.

## How to Start the App

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**Double-click** — it will show your local network IP address.

```
[START] Shift Scheduler running at:  
Local:    http://127.0.0.1:5000  
Network:  http://10.20.161.68:5000    ← use this on Android
```

### IMPORTANT

Your PC and Android phone must be on the **same Wi-Fi network**. Share the Network URL with your manager — they can open it in any browser.

## Login Accounts

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| Username | Password | Access Level                           |
|----------|----------|----------------------------------------|
| admin    | admin123 | Full access (generate, edit employees) |
| manager  | mgr456   | Generate schedules + view              |
| viewer   | view789  | View & download only                   |

### TIP

Your manager can use the `viewer` or `manager` account on their phone/PC.

## How to Generate a Schedule

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1. Log in → click  **Generate New Schedule**
2. Select **Year** and **Month**
3. **Step 2 – New Joiners** (optional): Click "Add New Joiner", fill in:
  - 6-digit Employee ID
  - Full Name
  - Role (Engineer / Operator / Technician / Apprentice)
  - Section (BA1, BA2, Engineering, etc.)
4. **Step 3 – Leavers** (optional): Enter comma-separated Employee IDs
5. **Step 4 – Leave Requests** (optional): Click "Add Leave Request", select employee and enter days (comma-separated)
6. Click  **Generate Schedule**

## Scheduling Rules (Applied Automatically)

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| Rule                             | Implementation                                                                     |
|----------------------------------|------------------------------------------------------------------------------------|
| ≥1 Operator per shift            | Checked every day for all 3 shifts; employees moved from W-off if needed           |
| ≥1 Support person per shift      | Engineers/Technicians redistributed to cover gaps                                  |
| Engineers → Shift 1 preferred    | Engineers assigned Shift 1 by default; only moved if critical                      |
| ≥2 week-offs around shift change | Extra W days inserted before each shift rotation                                   |
| Alternate weekly offs            | 1st & 3rd Saturdays + all Sundays = mandatory offs                                 |
| Leave honoured unless critical   | Leaves cancelled only if no W-off employee available                               |
| Leave cancellation priority      | Person with most leaves in last 2 months cancelled first, with justification shown |

# File Structure

```
Shift Scheduler/
├─ START_SCHEDULER.bat      ← Double-click to start
├─ app/
│   ├── app.py              ← Flask web application
│   ├── scheduler_engine.py ← Scheduling algorithm
│   └─ templates/           ← HTML pages
│       ├── base.html
│       ├── login.html
│       ├── dashboard.html
│       ├── generate.html
│       ├── schedule.html
│       ├── employees.html
│       └─ history.html
└─ data/
    ├── employees.json      ← 23 employees seeded from Oct 2026
    ├── leave_history.json  ← Tracks leave counts per month
    └─ schedules/           ← Generated schedules (JSON)
        └─ 2026_11.json    ← Example: Nov 2026
```

## Employee Roster (Pre-Loaded from October 2026)

| Role       | Count | Names                                                                                                         |
|------------|-------|---------------------------------------------------------------------------------------------------------------|
| Engineer   | 2     | Krishna Kumar, Vivek Morariya                                                                                 |
| Operator   | 7     | Bhargav, Dhavalkumar, Kantibhai, Jignesh, Rajesh, Jaymin, Anilbhai                                            |
| Technician | 12    | Anil Singh, Krishnadas, Nikesh, Bharatkumar, Vishnu, Kishor, Mehul, Ashwin, Bhavesh, Manubhai, Dhaval, Maulik |
| Apprentice | 2     | Vanshraj Makwana, Harsh Zala                                                                                  |

## Android "Add to Home Screen"

In Chrome on your Android:

1. Open `http://<PC_IP>:5000`

2. Tap : menu → **Add to Home Screen**

3. It installs as an app icon (PWA)

## Schedule View Features

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-  **Colour-coded table:** Green = Shift 1, Blue = Shift 2, Red = Shift 3, Yellow = W-Off, Grey = Leave
-  **Daily count summary** at the bottom (how many in each shift per day)
-  **Warnings panel:** shows any constraints that needed to be overridden
-  **Leave Adjustments:** explains exactly whose leave was cancelled and why
-  **Download Excel** — same format as your original October 2026 file

### NOTE

The leave history page (  History in the nav) shows a table of leave days per employee per month — this is used automatically to decide whose leave to cancel when scheduling constraints arise.